

From (13) and (14) we then conclude that

$$\begin{aligned}
W(u) &= \alpha_1 \|\sqrt{1-\theta} u\|_{L^2(\Omega)}^2 + \alpha_2 W_{0,\theta}(u) \\
&\leq \alpha_1 \liminf_{n \rightarrow \infty} \|\sqrt{1-\theta} u_n\|_{L^2(\Omega)}^2 + \alpha_2 \liminf_{n \rightarrow \infty} W_{0,\theta}(u_n) \\
&\leq \liminf_{n \rightarrow \infty} \left(\alpha_1 \|\sqrt{1-\theta} u_n\|_{L^2(\Omega)}^2 + \alpha_2 W_{0,\theta}(u_n) \right) \\
&= \liminf_{n \rightarrow \infty} W(u_n),
\end{aligned}$$

what proves (H2).

To prove (H3) let $\{u_n\} \subset L^2(\Omega)$ be such that $W(u_n) \leq c_1 < \infty$, $\forall n$. We want to show that there exist $\{u_{n_j}\} \subset \{u_n\}$ and $u \in L^2(\Omega)$ such that $u_{n_j} \xrightarrow{w} u$. For this note that

$$(1 - \varepsilon_2) \|u_n\|_{L^2(\Omega)}^2 \leq \|\sqrt{1-\theta} u_n\|_{L^2(\Omega)}^2 \leq W(u_n) \leq c_1. \quad (15)$$

Thus $\|u_n\|_{L^2(\Omega)}$ is uniformly bounded and therefore there exist $\{u_{n_j}\} \subset \{u_n\}$ and $u^* \in L^2(\Omega)$ such that $u_{n_j} \xrightarrow{w} u^*$. Hence, by Theorem 1.3, the functional $F_\theta(u)$ given by (12) has a global minimizer $u^* \in L^2(\Omega)$. The condition $\theta(x) \leq \varepsilon_2 < 1$ for a.e. $x \in \Omega$ clearly implies the strict convexity of F_θ and therefore the uniqueness of such a global minimizer.

To prove the second part of the theorem, assume further that there exists $\varepsilon_1 > 0$ such that $\theta(x) \geq \varepsilon_1$ for a.e. $x \in \Omega$. Following the proof of Theorem 5.1 in [11], it suffices to show that under this additional hypothesis the weak limit u in (H3) above belongs to $BV(\Omega)$. For this note that from (15) it follows that there exist $c_2 < \infty$ such that

$$\|u_n\|_{L^1(\Omega)} \leq c_2 \quad \forall n. \quad (16)$$

Also, by Theorem 2.4 $W_{0,\theta}(u) \geq \varepsilon_1 J_0(u) \quad \forall u \in \mathcal{M}(\Omega)$. This, together with (16) implies that

$$\|u_n\|_{BV(\Omega)} = \|u_n\|_{L^1(\Omega)} + J_0(u_n) \leq c_2 + \frac{W_{0,\theta}(u_n)}{\varepsilon_1} \leq c_3 < \infty \quad \forall n,$$

where the previous to last inequality follows from the uniform boundedness of $W_{0,\theta}(u_n)$, which, in turn, follows from the uniform boundedness of $W(u_n)$. Hence the fact that the weak limit in (H3) is in $BV(\Omega)$ follows from the compact embedding of $BV(\Omega)$ in $L^2(\Omega)$. This result is an extension of the Rellich-Kondrachov Theorem which can be found, for instance, in [2] and [3]. $\square$

Remark 2.7. Note that if $\theta(x) = 0 \quad \forall x \in \Omega$, then $W(u) = \|u\|_{L^2(\Omega)}^2$ and F_θ as defined in (12) is the classical Tikhonov-Phillips functional of order zero. On the other hand, if $\theta(x) = 1 \quad \forall x \in \Omega$ then $W(u) = J_0(u)$ and F_θ has a global minimizer provided that $T\chi_\Omega \neq 0$. If moreover T is injective then such a global minimizer is unique. All these facts follow immediately from Theorems 3.1 and 4.1 in [1].

Theorem 2.8. Let $\Omega \subset \mathbb{R}^n$ be a bounded open convex set with Lipschitz boundary, $\mathcal{X} = L^2(\Omega)$, $\mathcal{Y}$ a normed vector space, $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$, $v \in \mathcal{Y}$, α_1, α_2 positive constants and $\theta \in \widehat{\mathcal{M}}(\Omega)$ such that $\frac{1}{1-\theta} \in L^1(\Omega)$ and $\frac{1}{\theta} \in L^\infty(\Omega)$. Then the functional (12) has a unique global minimizer $u^* \in BV(\Omega)$.

Proof. Let us consider the functional

$$W(u) \doteq \alpha_1 \|\sqrt{1-\theta} u\|_{L^2(\Omega)}^2 + \alpha_2 W_{0,\theta}(u), \quad u \in L^2(\Omega).$$

By virtue of Theorems 1.3 and 2.6 and the compact embedding of $BV(\Omega)$ in $L^2(\Omega)$, it suffices to show that $W(\cdot)$ satisfies (H1) and (H2) and that every W -bounded sequence is also BV -bounded. Clearly $W(\cdot)$ satisfies (H1) with $\gamma = 0$. That it satisfies (H2) follows immediately from the fact that the condition $\frac{1}{1-\theta} \in L^1(\Omega)$ implies that $\|\sqrt{1-\theta} \cdot\|_{L^2(\Omega)}$ is a norm.

Now, let $\{u_n\} \subset L^2(\Omega)$ be a W -bounded sequence, i.e. such that $W(u_n) \leq c < \infty$, $\forall n$. We will show that $\{u_n\}$ is BV -bounded. Since $W(u_n)$ is uniformly bounded, there exist $K < \infty$ such that $\|\sqrt{1-\theta} u_n\|_{L^2(\Omega)} \leq K \forall n$. From this and the fact that $\frac{1}{1-\theta} \in L^1(\Omega)$ it follows that

$$\begin{aligned} \|u_n\|_{L^1(\Omega)} &= \int_{\Omega} \frac{1}{\sqrt{1-\theta}} \sqrt{1-\theta} |u_n| dx \\ &\leq \left(\int_{\Omega} \frac{1}{1-\theta} dx \right)^{\frac{1}{2}} \left(\int_{\Omega} (1-\theta) u_n^2 dx \right)^{\frac{1}{2}} \\ &= \left\| \frac{1}{1-\theta} \right\|_{L^1(\Omega)}^{\frac{1}{2}} \|\sqrt{1-\theta} u_n\|_{L^2(\Omega)} \\ &\leq K \left\| \frac{1}{1-\theta} \right\|_{L^1(\Omega)}^{\frac{1}{2}} < \infty \quad \forall n. \end{aligned} \tag{17}$$

On the other hand from Theorem 2.4 $J_0(u) \leq W_{0,\theta}(u) \left\| \frac{1}{\theta} \right\|_{L^\infty(\Omega)} \forall u \in L^2(\Omega)$. Since $\frac{1}{\theta} \in L^\infty(\Omega)$ and $W_{0,\theta}(u_n)$ is uniformly bounded, it then follows that there exists $C < \infty$ such that

$$J_0(u_n) \leq C \forall n. \tag{18}$$

From (17) and (18) it follows that

$$\|u_n\|_{BV(\Omega)} = \|u_n\|_{L^1(\Omega)} + J_0(u_n) \leq K \left\| \frac{1}{1-\theta} \right\|_{L^1(\Omega)}^{\frac{1}{2}} + C < \infty \forall n.$$

Hence $\{u_n\}$ is BV -bounded. The existence of a global minimizer of functional (12) belonging to $BV(\Omega)$ then follows. Finally note that the condition $\frac{1}{1-\theta} \in L^1(\Omega)$ implies the strict convexity of F_θ and therefore the uniqueness of the global minimizer. $\square$

Remark 2.9. Note that the condition $\frac{1}{1-\theta} \in L^1(\Omega)$ in Theorem 2.8 is weaker than the condition $\theta(x) \leq \varepsilon_2 < 1$ for a.e. $x \in \Omega$ of Theorem 2.6. While the latter suffices to guarantee the existence of a global minimizer in $L^2(\Omega)$, the former does not. However this weaker condition $\frac{1}{1-\theta} \in L^1(\Omega)$ together with the condition $\frac{1}{\theta} \in L^\infty(\Omega)$ are enough for guaranteeing not only the existence of a unique global minimizer, but also the fact that such a minimizer belongs to $BV(\Omega)$.

It is timely to note that in both Theorems 2.6 and 2.8, the function θ cannot assume the extreme values 0 and 1 on a set of positive measure. In some cases a pure BV regularization in some regions and a pure L^2 regularization in others may be desired, and therefore that restraint on the function θ will turn out to be inappropriate. In the next three theorems we introduce different conditions which allow the function θ to take the extreme values on sets of positive measure.